

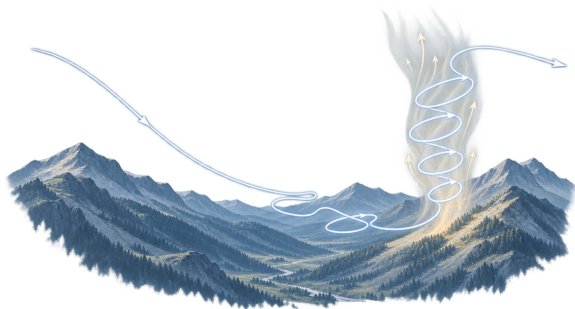
INTERNSHIP & THESIS ROADMAP

From flight logs to optimal paths

Possible research paths
and timeline

Matteo Di Sante

September 2026 – February 2027



Econophysics Lab

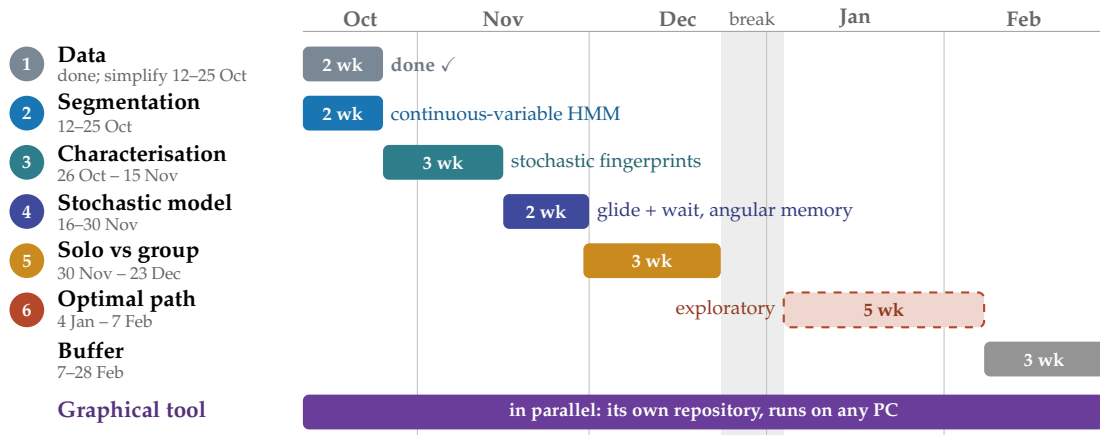
CFM

ILB Institut
Louis
Bachelier

The plan at a glance

12 Oct 2026 – 28 Feb 2027

2



Buffer: absorb delays, or go deeper on the most interesting results

Download

186,052

paraglider tracklogs

6,716

hang-glider tracklogs

preprocessing
→

Clean archive

156,406

paraglider flights (84.1%)

6,094

hang-glider flights (90.7%)

Simplify the preprocessing

NEXT

Today's version has too many parameters, is too complex and discards too many flights (paragliders: 16%).

In parallel with segmentation, 12–25 Oct.

Vilpellet's HMM

X^{v_z} : climbing?

X^{curv} : turning one way?

X^{str} : straight?

yes/no, on a rolling 30 s window

**Checked in the viewer:
works well, but misses
some climbs**

Only for 1 Hz



30%

paragliders

72%

hang gliders

**of flights left out: loggers
slower than 1 Hz**

Continuous HMM



vertical speed

turn rate

turn coherence

$$\begin{aligned} \langle \dot{z} \rangle_W \\ \langle |\dot{\theta}| \rangle_W \\ \frac{|\int_W \dot{\theta} dt|}{\int_W |\dot{\theta}| dt} \end{aligned}$$

W: time window

**To validate:
downsampled 1 Hz flights
and/or hand-labelled
flights**

Goal: phase labels for all flight time, whatever the logger



$$\text{MSD} \sim \tau^{2H}$$

$H = 0.869$ paragliders
 $H = 0.841$ hang gliders
 3% apart; Vilpellet: 0.88

first look



Anisotropy

How do ridges and wind
 shape the anisotropy?

first look



Monofractal?

$\langle |\Delta x|^q \rangle \sim \tau^{\zeta(q)}$: is
 $\zeta(q) = qH$?

first look



Self-similar?

Do the rescaled laws
 $\tau^H P(x, \tau)$ collapse?

first look



Memory

Autocorrelation of
 velocity



Open vs closed

Two classes of motion, to
 treat separately

to explore



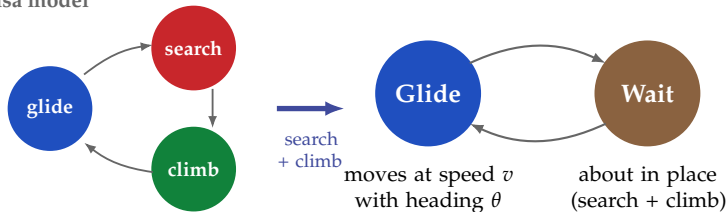
Terrain, circuit, wing

How terrain, circuit type and wing class affect
 the motion.

Other tools?

Which other descriptors of stochastic motion
 fit our case?

Pisa model

**Memory decides long times**

$$\text{MSD}(\tau) = 2 \int_0^\tau (\tau - s) C_v(s) ds$$

$$\int_0^\infty C_v ds < \infty: \text{diffusive}$$

$$C_v \sim s^{-\gamma}, \gamma < 1: \text{MSD} \sim \tau^{2-\gamma}$$

1 Heading data

$C_\theta(\tau)$ within and across glides

2 Angular law

an equation for $\theta(t)$ compatible with the observed data

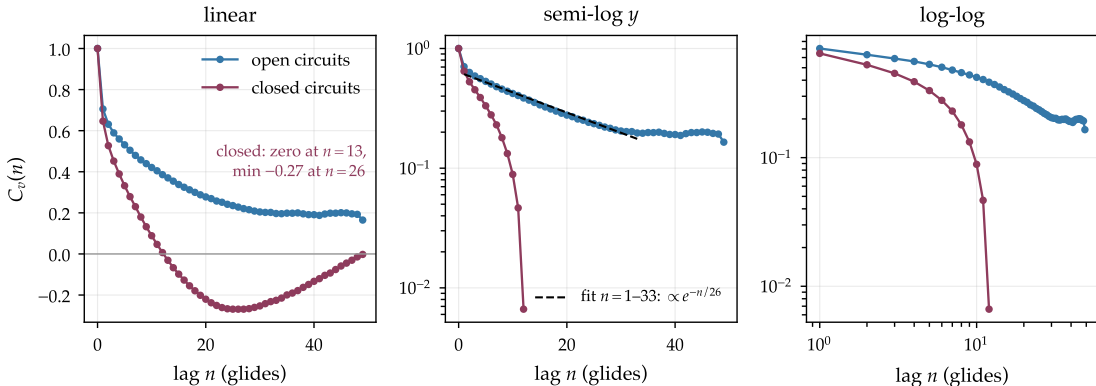
3 Long times

asymptotic superdiffusion?

4 Check

not only H : the other features measured in step 3

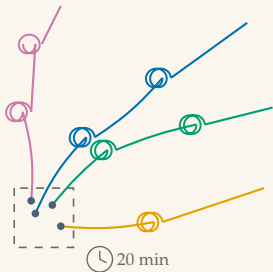
$\mathbf{v}_k = \Delta \mathbf{r}_k / T_k$: mean velocity vector of glide k . $C_v(n) = \langle \mathbf{v}_k \cdot \mathbf{v}_{k+n} \rangle / \langle |\mathbf{v}_k|^2 \rangle$



Open vs closed: very different motion from the first lags.

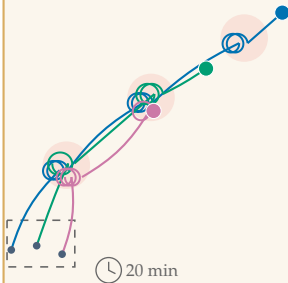
Open: $C_v \approx 0.63 e^{-n/26}$ up to $n = 33$: diffusive at long times?

Same take-off
cell, same 20 min



same start, different paths

Flying together



same thermals

1

An operational definition

Same take-off cell +
same departure window
does not imply group flight



2

Then the analysis

solo vs group: where they agree
and where they differ

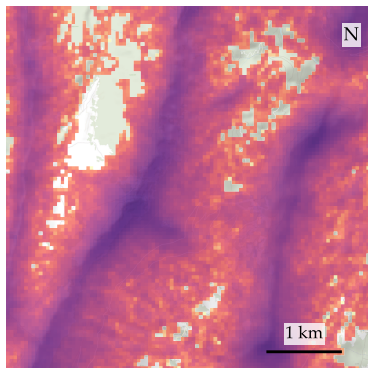
Then compare with step 3:

MSD and H

scaling

memory

phase durations



Thermal hours/km² (log)

Chartreuse, 5 × 5 km.
All years pooled.

1

Data on glider climb segments

→ a thermal distribution $P(x, y)$

Several definitions, depending on the research question.



2

Updraft distribution $p(w \mid \text{site})$

w : vertical air velocity

3

How do both distributions change?

between terrain classes: mountains, hills, plains

within one class: from one mountain (or plain) to another



4

Optimal 2D path for each terrain type,
using the thermal distributions from the data



5

Compare optimal paths with recorded flights



Today

Works, but **not reviewed or tested**

Mixed with the analysis code, so it **runs only on my Mac**

Short term: today's viewer

- own repository, runs on any PC
- no new work on it

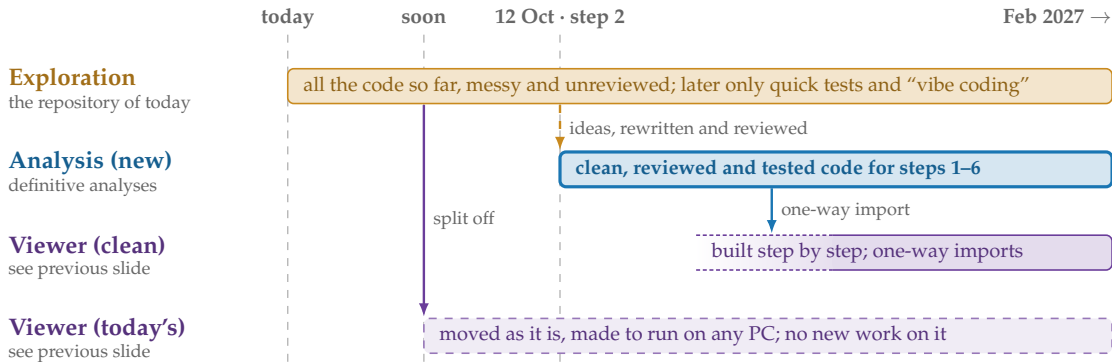
Usable soon, still unreviewed

In parallel: a clean viewer

- reviewed, one-way imports, any PC
- all new features go here

Trustworthy, but it takes longer

Make today's viewer usable by everyone, but invest the time only in the clean viewer



Explore fast in one place; build the internship results only on reviewed code